

Customer Journey Map

Date	28 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	Stage 1 Data Collection & Setup	Stage 2 Model Training & Prediction	Stage 3 Flask Web Application
Actions <i>What does the customer do?</i>	- Downloads HDI dataset from Kaggle (195 countries, 82 features) - Sets up Python environment with Anaconda - Installs required libraries using pip	- Runs Jupyter Notebook for data preprocessing - Trains Linear Regression model on HDI data - Saves trained model as HDI.pkl using Pickle	- Opens Flask web app in browser - Enters HDI indicators in input form - Views predicted HDI category and score on result page
Touchpoint <i>What part of the service do they interact with?</i>	- Kaggle dataset repository - Anaconda Navigator / VS Code terminal - pip install commands	- Jupyter Notebook (HumDevIndex.ipynb) - Scikit-learn Linear Regression model - Pickle model serialization (HDI.pkl)	- Flask home page (home.html) - HDI input form (indexnew.html) - Prediction result page (resultnew.html)
Customer Thought <i>What is the customer thinking?</i>	- Will this dataset have all required HDI indicators? - Are all libraries compatible with Python 3.12? - How do I handle 18 missing values across columns?	- Which features have strongest correlation with HDI? - Is Linear Regression the right algorithm? - How accurate will the R-squared score be?	- Is the web interface easy to use? - Are the HDI categories accurately classified? - Can this tool help with real policy decisions?
Customer Feeling <i>What is the customer feeling?</i>	- Curious but slightly overwhelmed by 82 columns - Motivated to explore the HDI indicators - Relieved when environment setup completes	- Excited to see model training results - Hopeful that R-squared score will be high - Satisfied when model predictions look accurate	- Impressed by the clean Flask interface - Confident in the HDI prediction results - Empowered to make data-driven development decisions
Process Ownership <i>Who is in the lead on this?</i>	- Pechetti Lilly Lakshmi — Reading & Understanding Dataset - Spandhana	Gara Bhandhavi — Data Visualization, Preprocessing, Null Values	Balla Kamala Anjali — Flask Backend, Running Application, Conclusion

	Mondam — Environment Setup, Libraries, Folder Structure	- Spandhana Mondam — Train/Test Split, Model Training & Prediction - Bhagyalatha Kadali — Saving the Model (HDI.pkl)	Spandhana Mondam — Project Workflow, ER Diagram, Links
Opportunities <i>How can we improve this stage?</i>	- Add automated data cleaning pipeline - Include more recent UNDP HDI datasets - Add data validation before preprocessing - introducing dashboard contains details of all countries HDI score	- Try Random Forest for better accuracy - Add cross-validation for robust evaluation - Include more HDI indicators as features	- Add data visualization charts on result page - Enable comparison of multiple countries - Deploy Flask app to cloud for public access